Determining the Size and Stiffness of Objects Using Underactuated Grasping

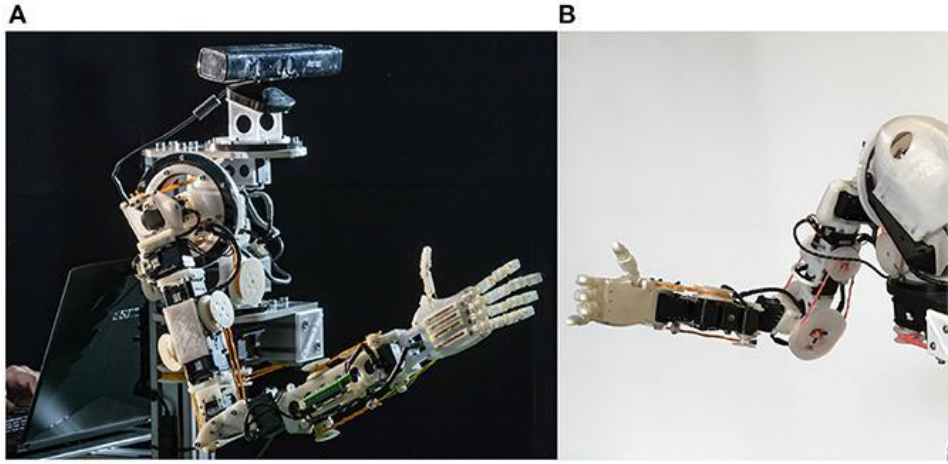
Matthew Hunter

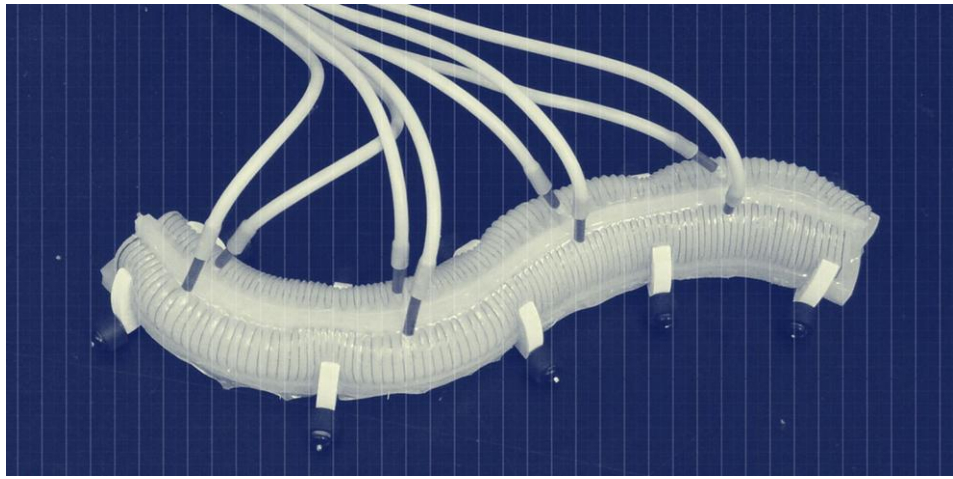
PI: Dr. Hammond

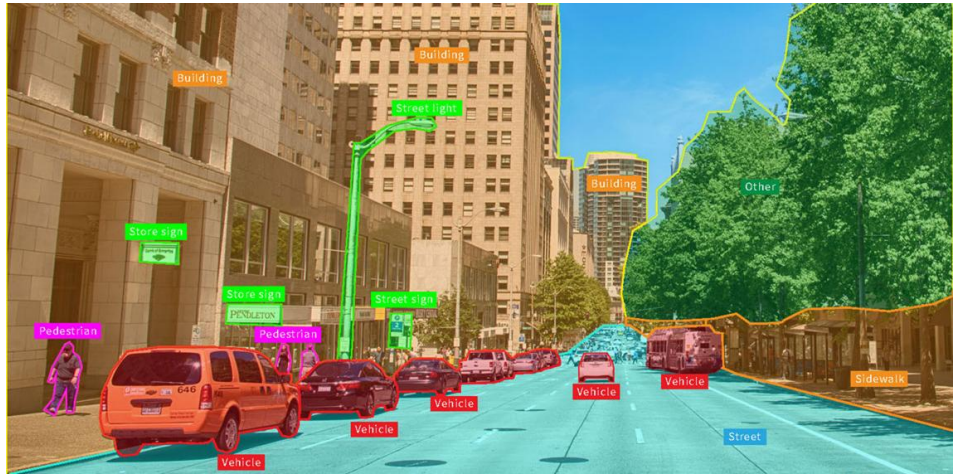
Graduate Mentor: Houriyeh Majditehran

Background

- Machines are implemented to perform tasks humans cannot
- Machines don't have "Nervous system"
- Through haptic feedback touch sensation can be transferred from the environment, through a machine and into humans







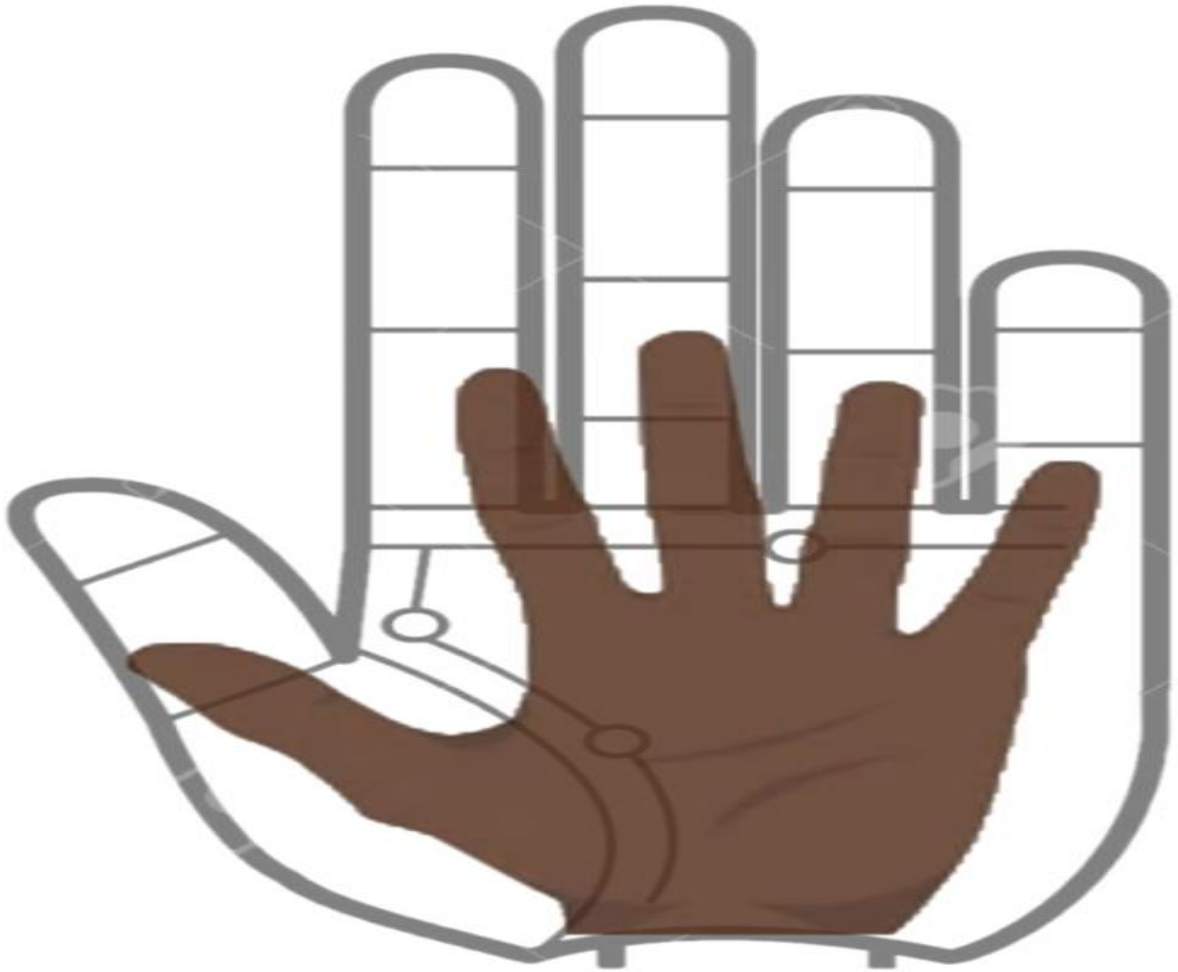
Motivation

Technology has implementation in

- **Construction**
- **Surgery**
- **Prosthetics**

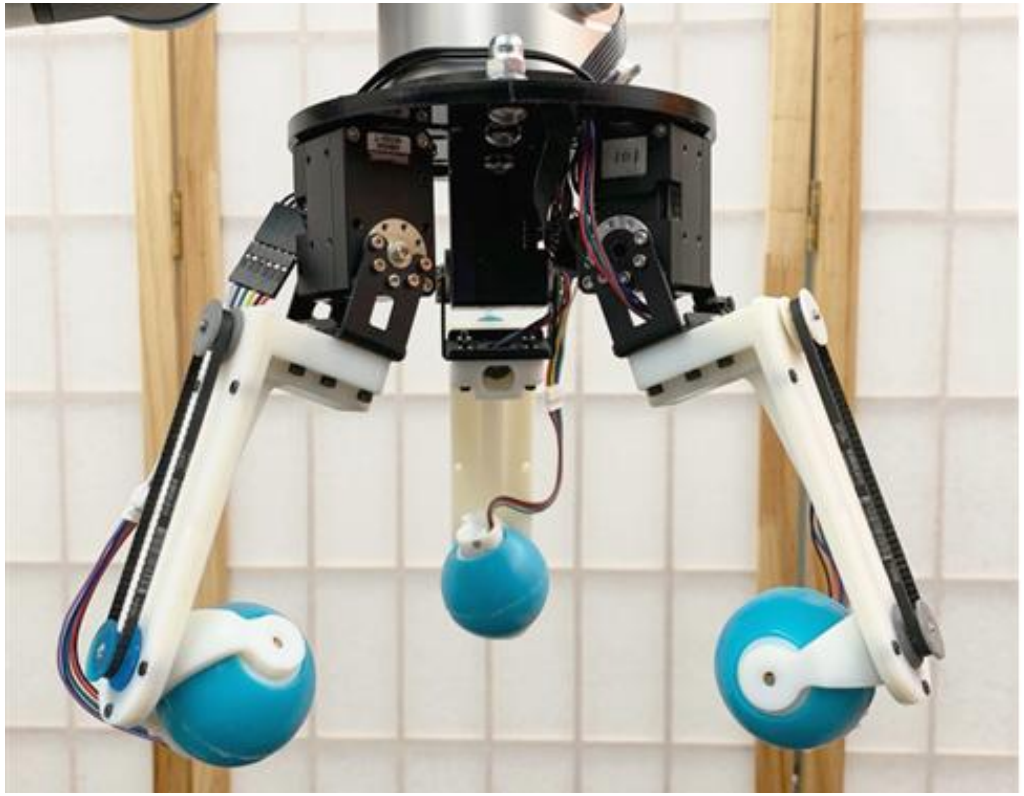
Problem Statement

Using an under-actuated grasper, we want to be able to distinguish between objects of different sizes and stiffnesses, then map touch sensation to a human

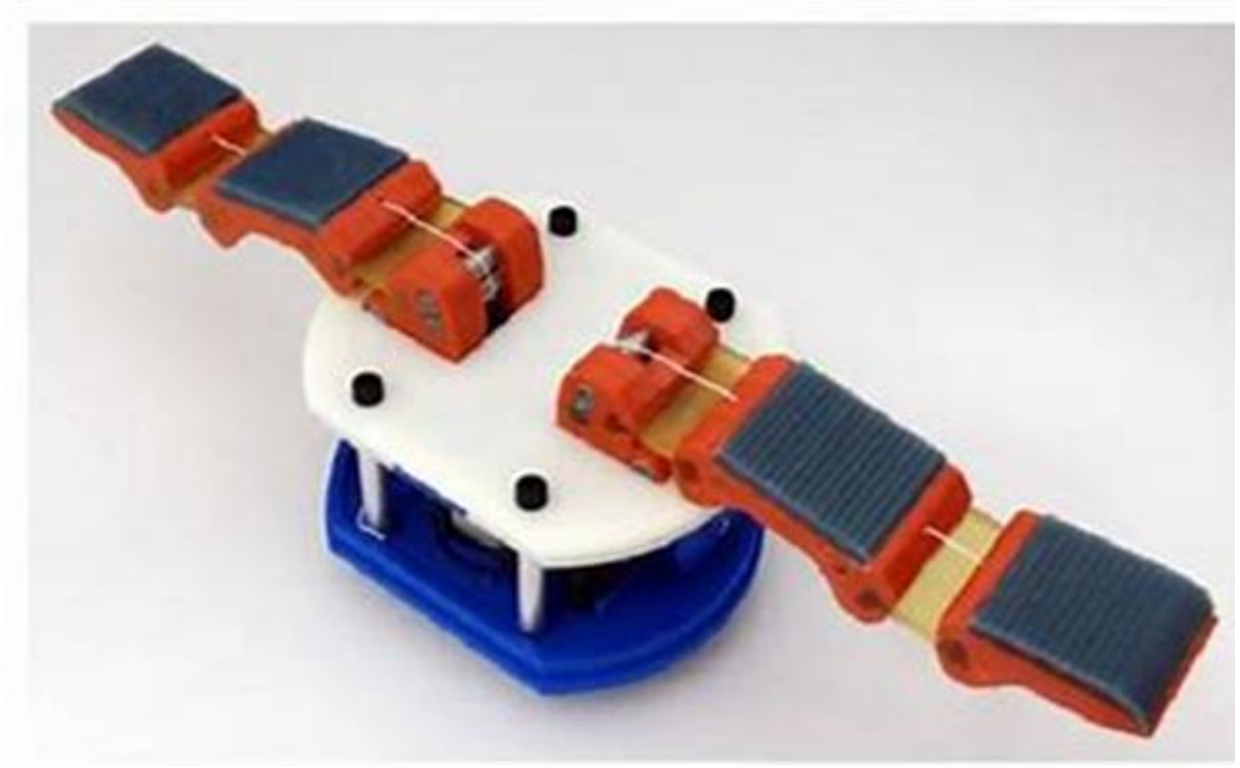


Fully Actuated vs Under Actuated

Fully Actuated

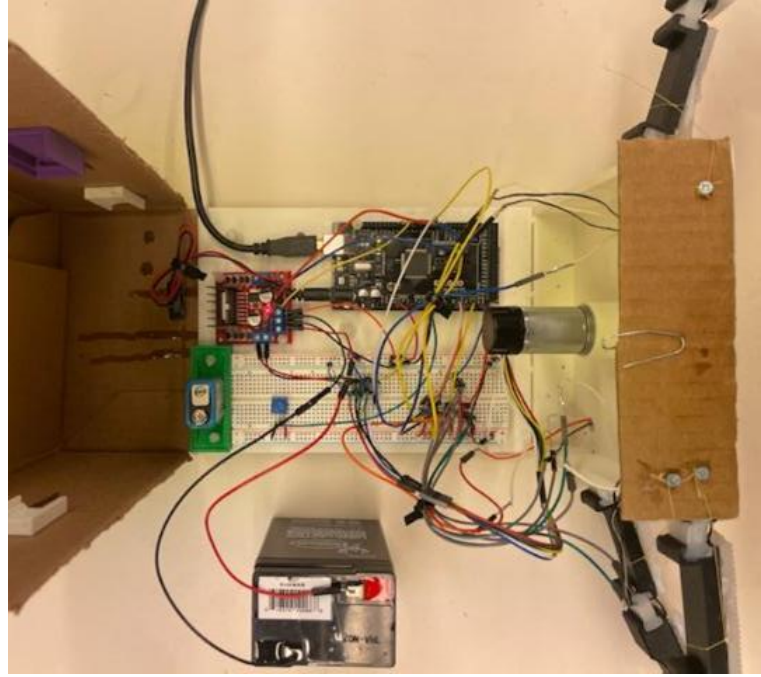


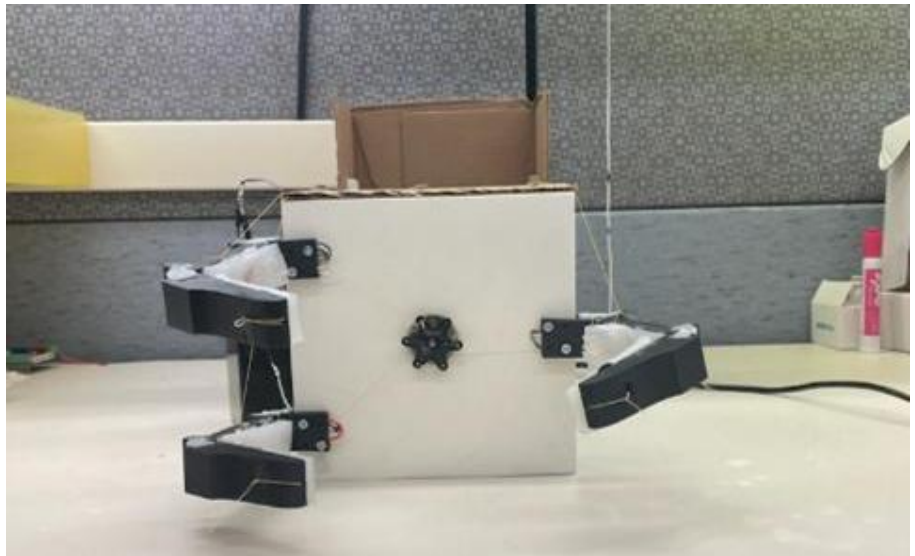
Under Actuated

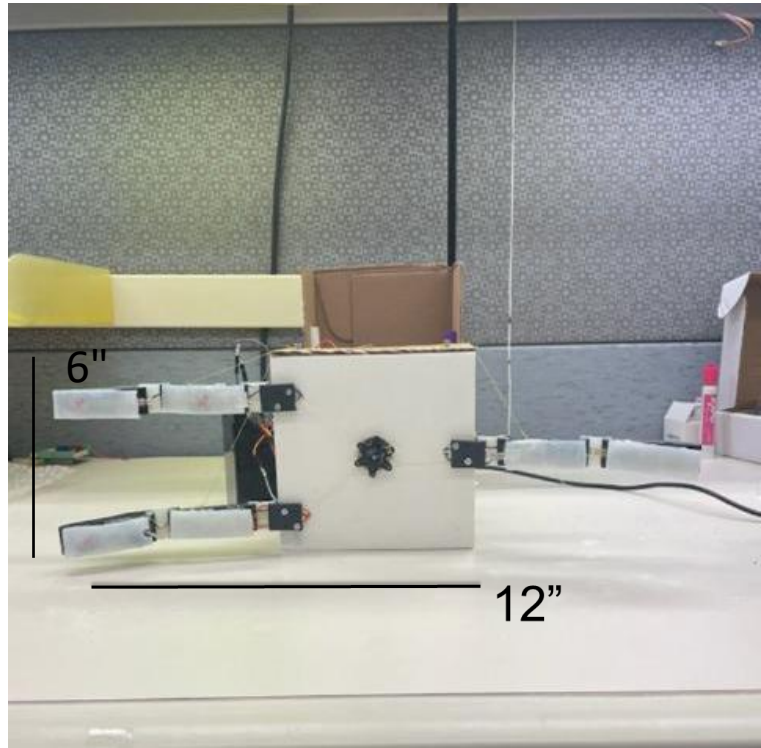


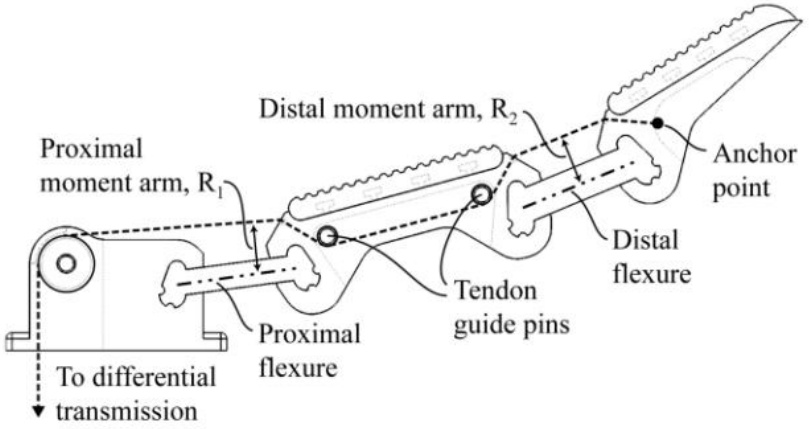
Under Actuated Grasper Design

- Hand length 12"
- Hand breadth 6"
- Weight 1.68 lbs or .76 kg









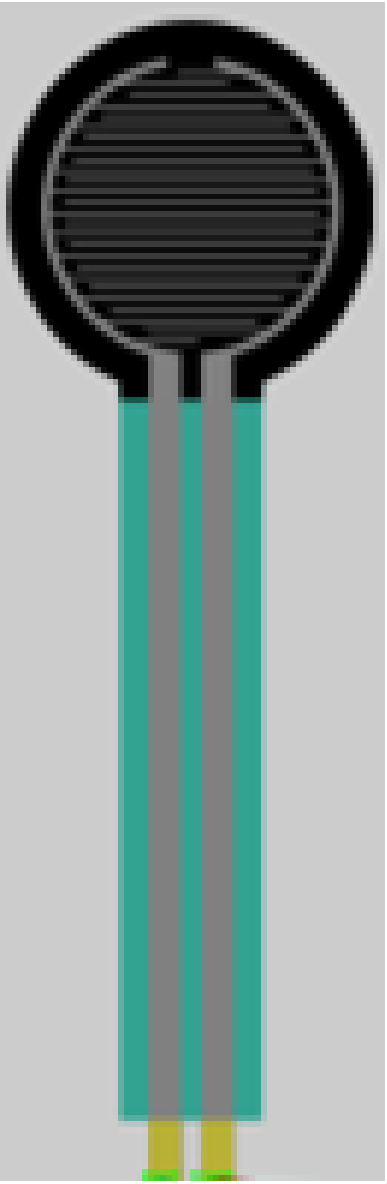
7"

Under Actuated Grasper Design

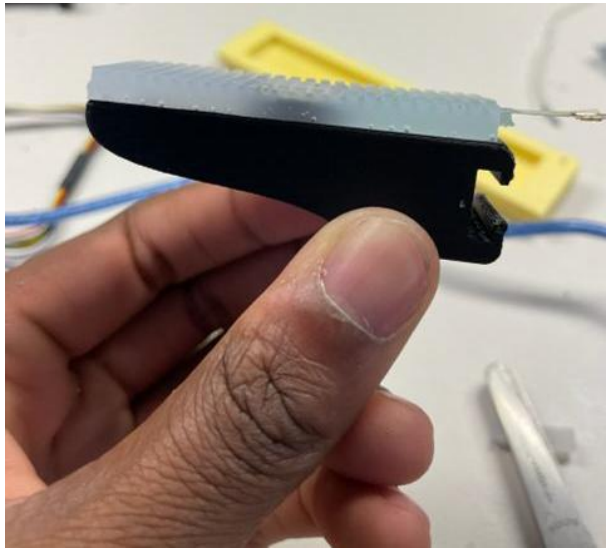
- Three segments
- Silicone joints
- Slots for actuation strings and wires
- Rounded edges to increase ROM

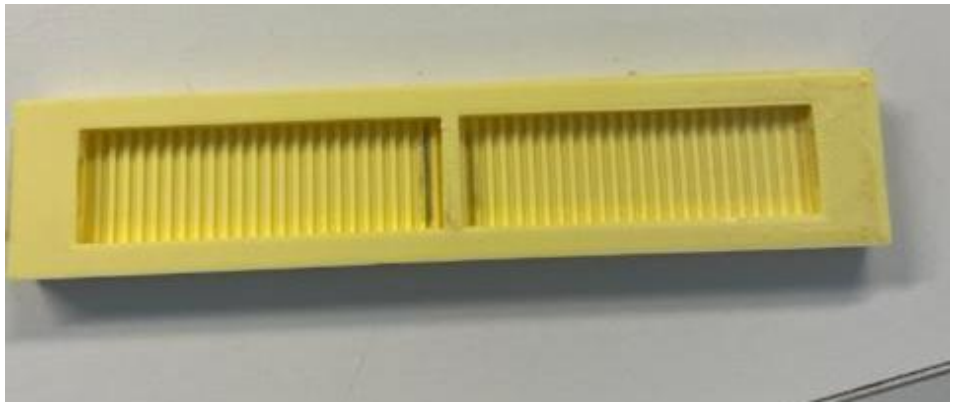
Force Sensitive Resistor (FSR)

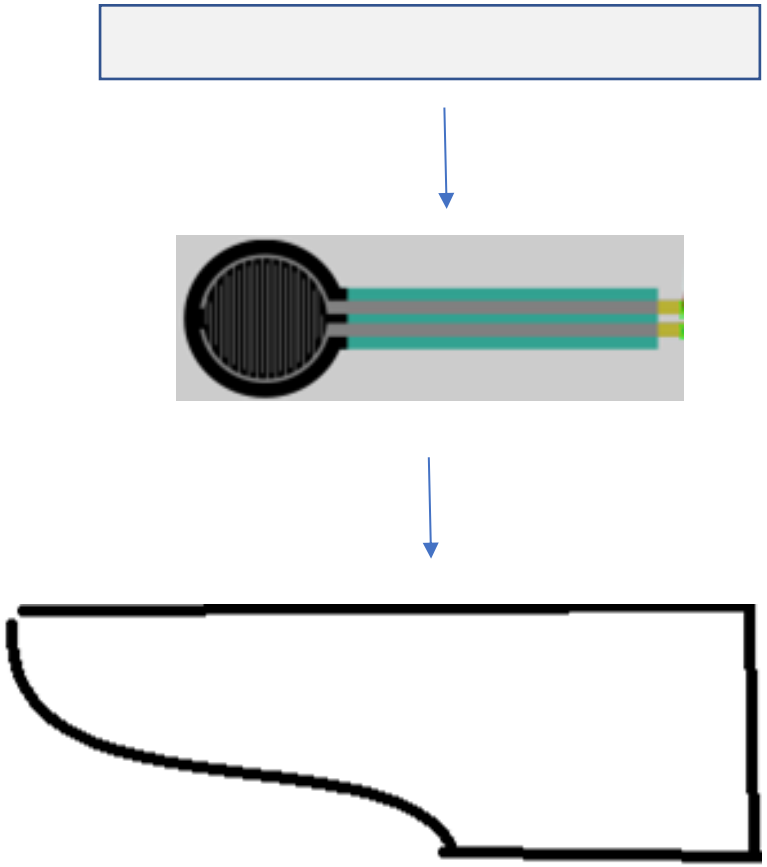
- Pressure changes the resistance of the component with voltage proportional to pressure
- Good for measuring relative pressure instead of exact unit measurements
- Similar to a potentiometer



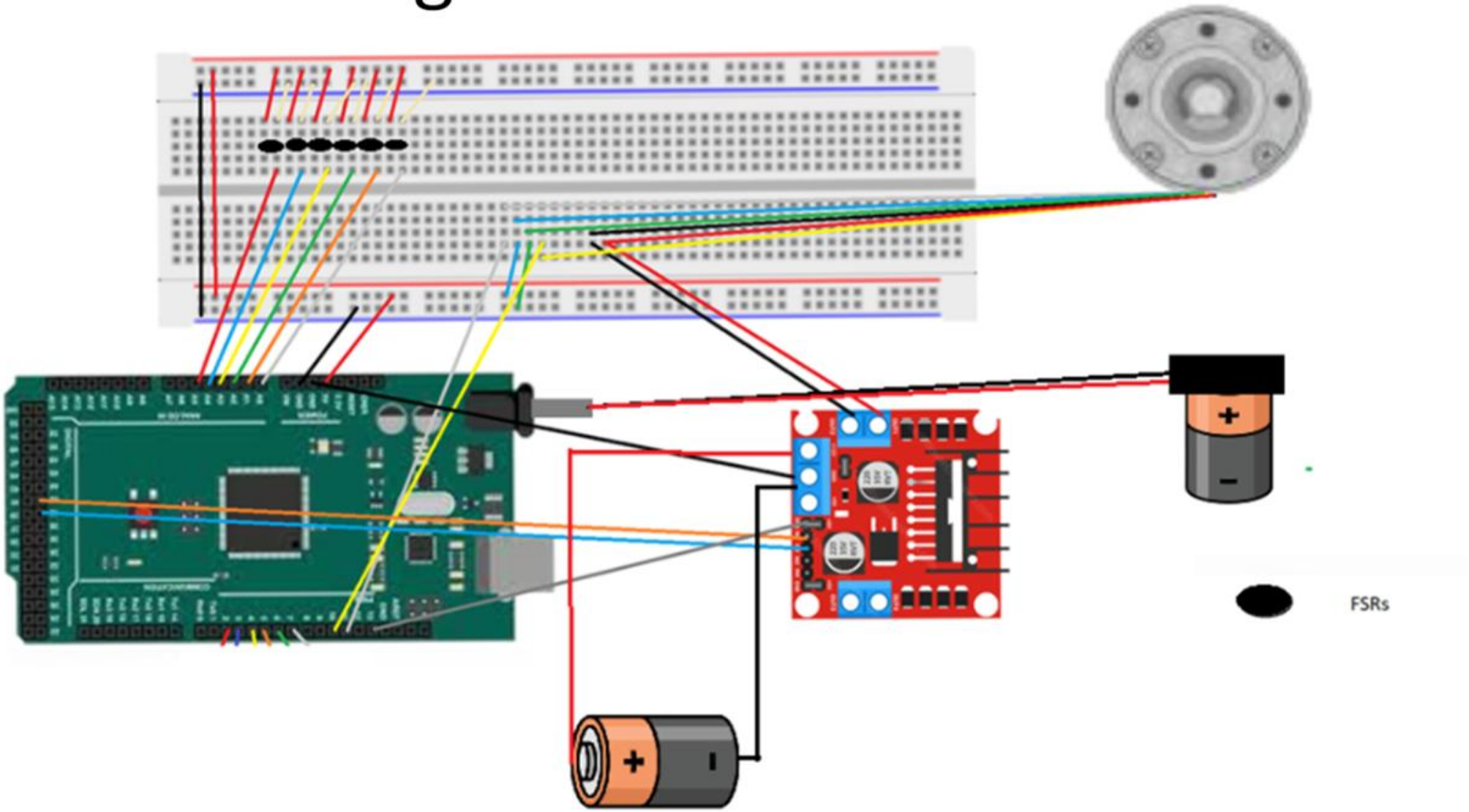
Sensor Implementation



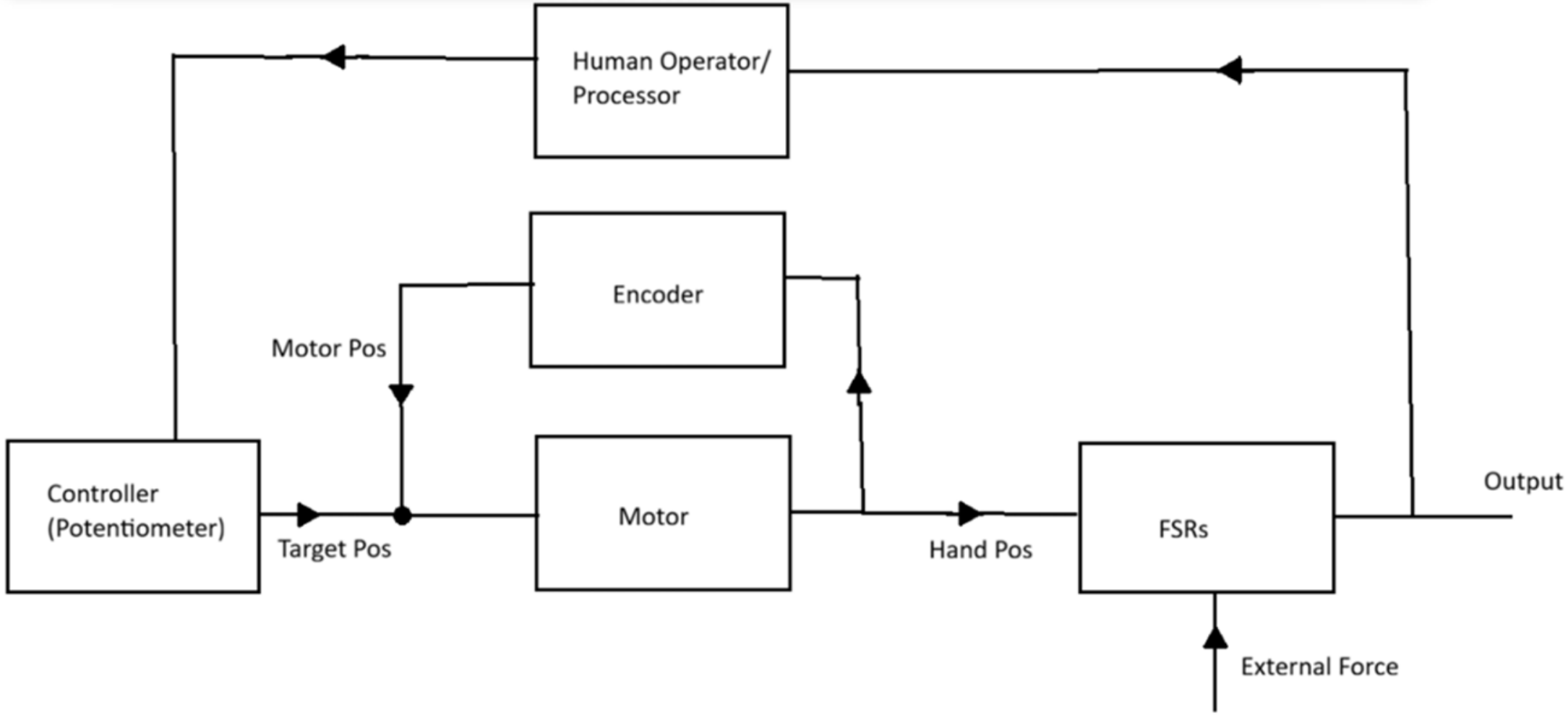


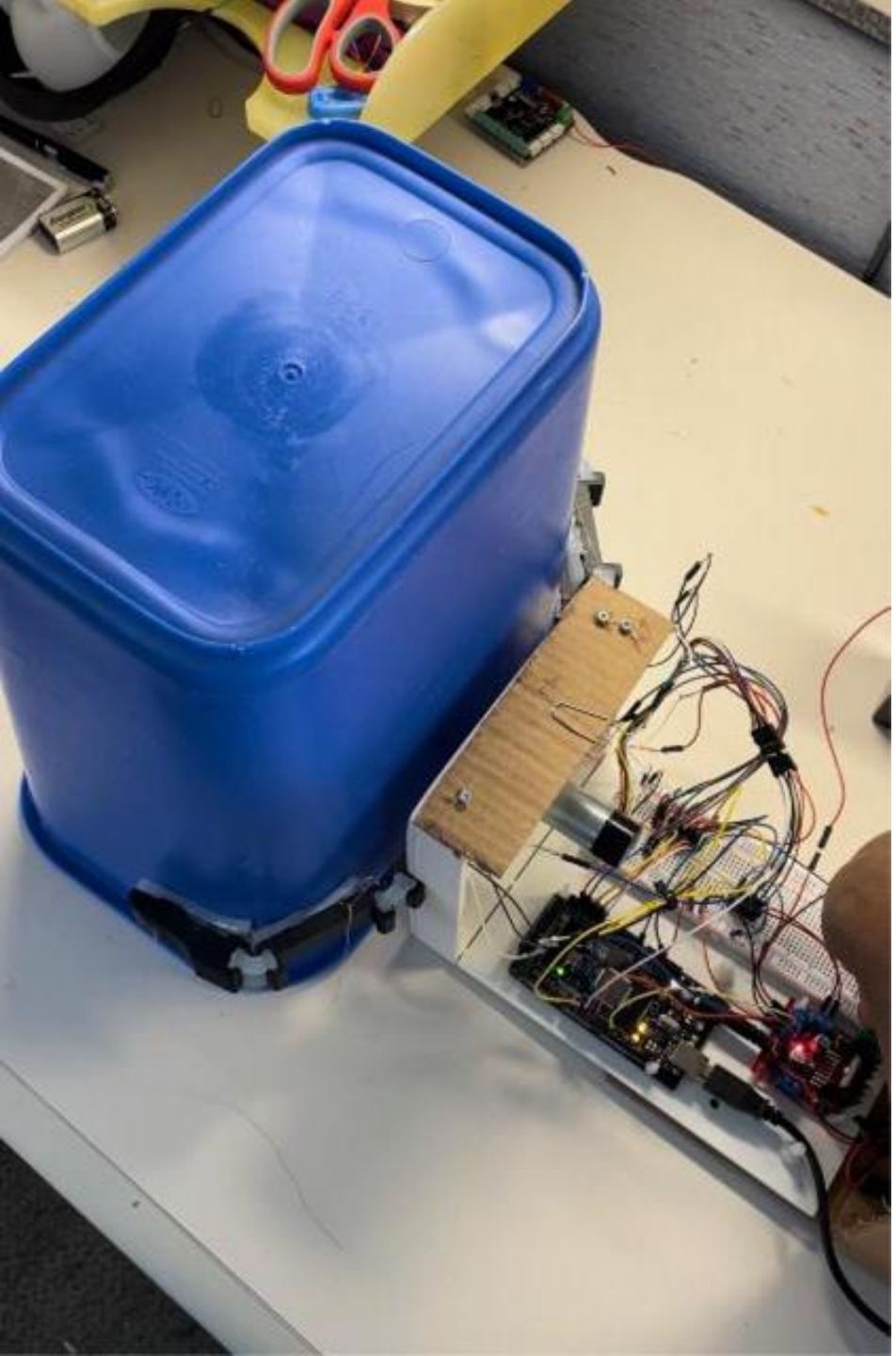


Electrical Diagram



Control Algorithm



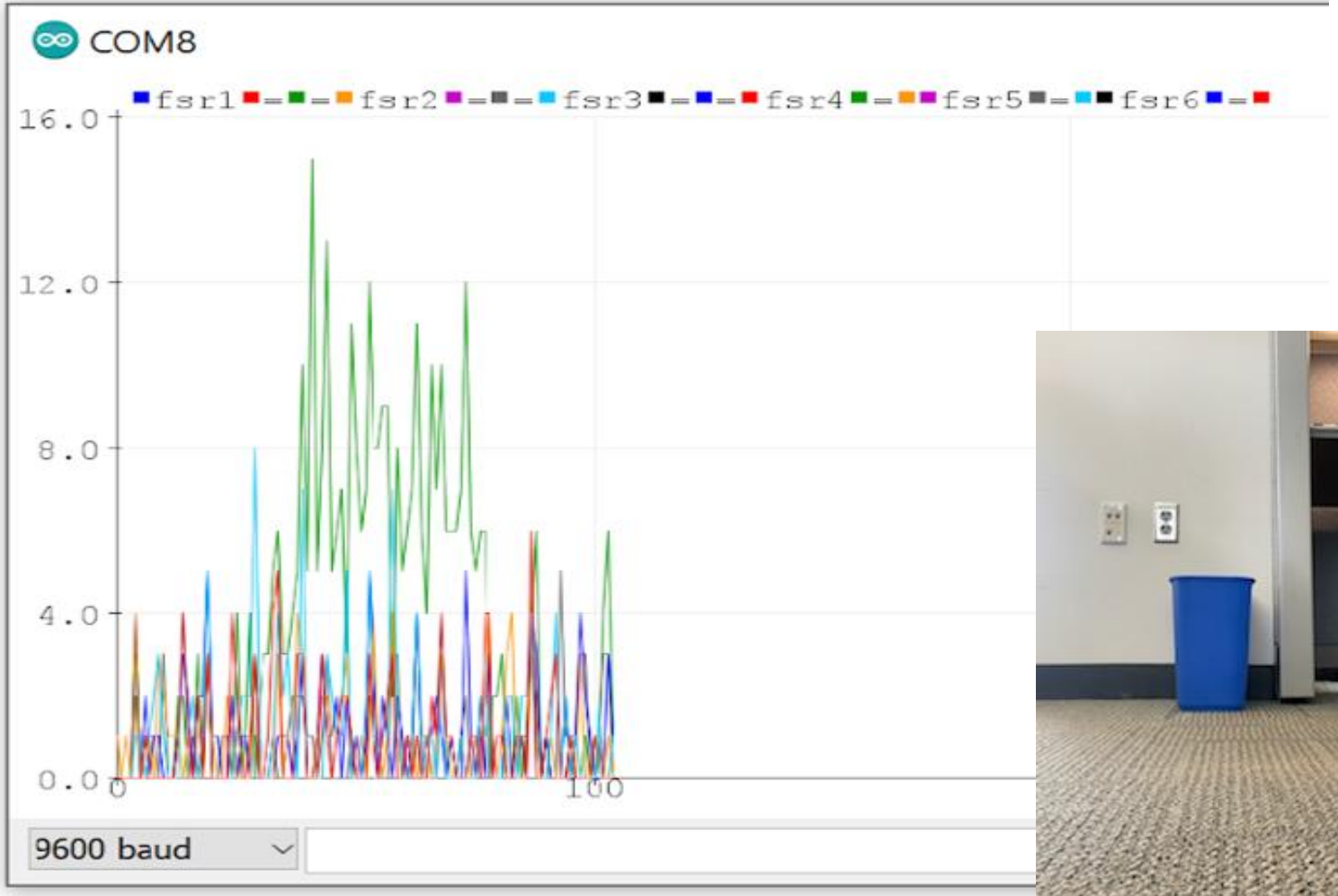


Trial

- Used three rounded prisms
- Each has a long and short side
- Each has a different stiffness
- Six sizes, three Stiffnesses

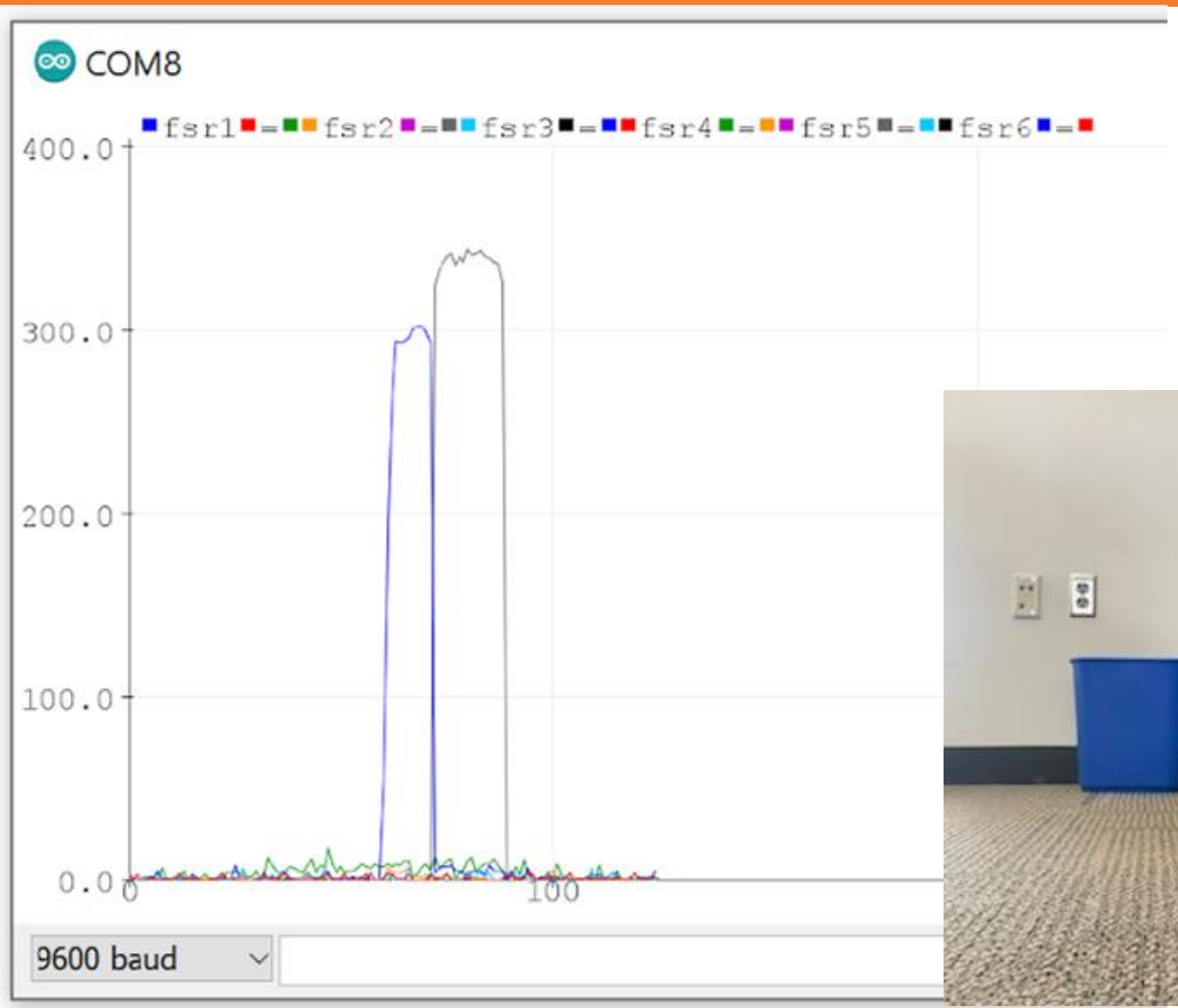
Results

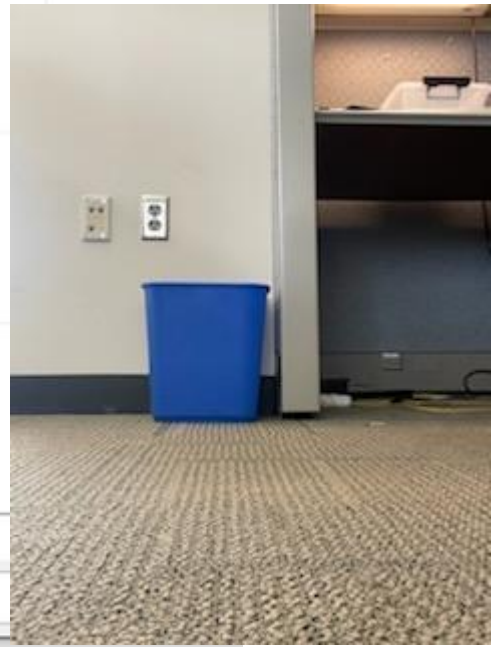
FSR 1
FSR 2
FSR 3
FSR 4
FSR 5
FSR 6


COM8
fsr1 fsr2 fsr3 fsr4 fsr5 fsr6
16.0
12.0
8.0
4.0
0.0
0
100
9600 baud

Results

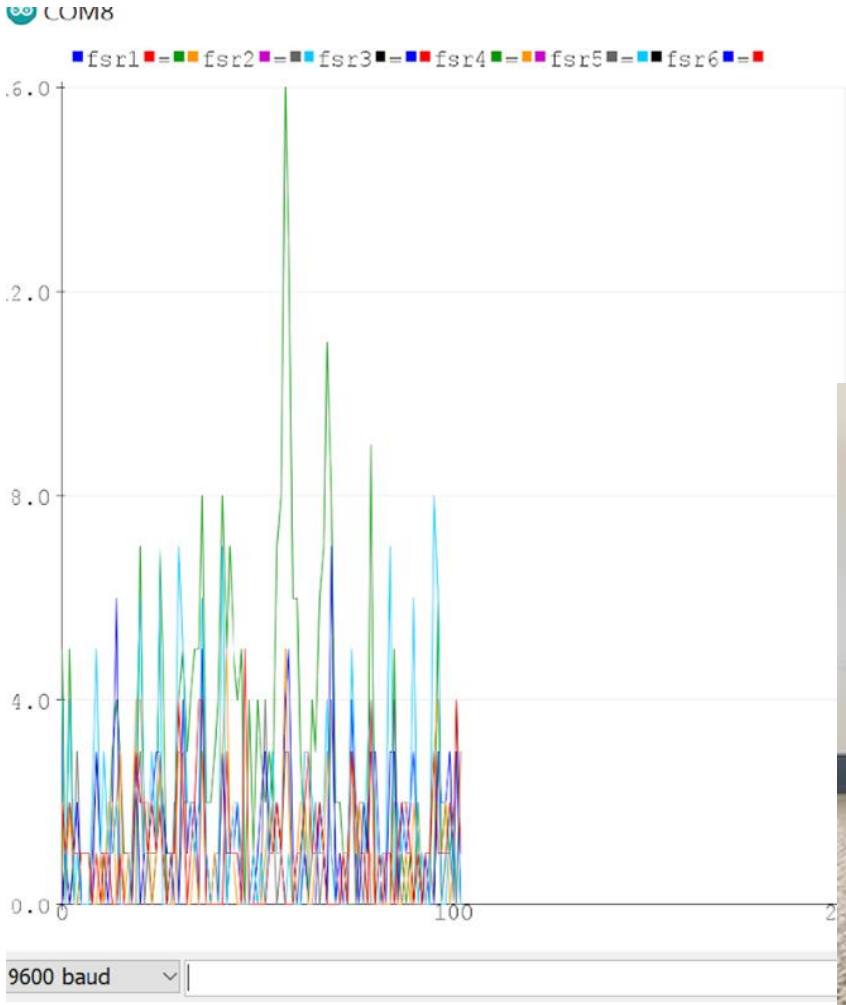
FSR 1
FSR 2
FSR 3
FSR 4
FSR 5
FSR 6

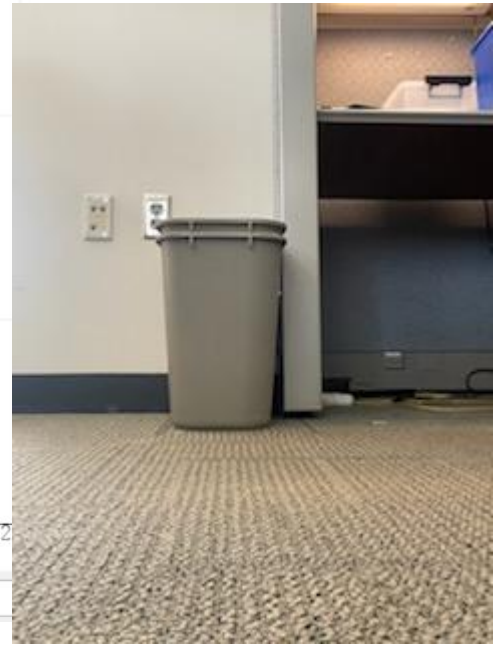

COM8
fsr1 fsr2 fsr3 fsr4 fsr5 fsr6
400.0
300.0
200.0
100.0
0.0
0
100
9600 baud



Results

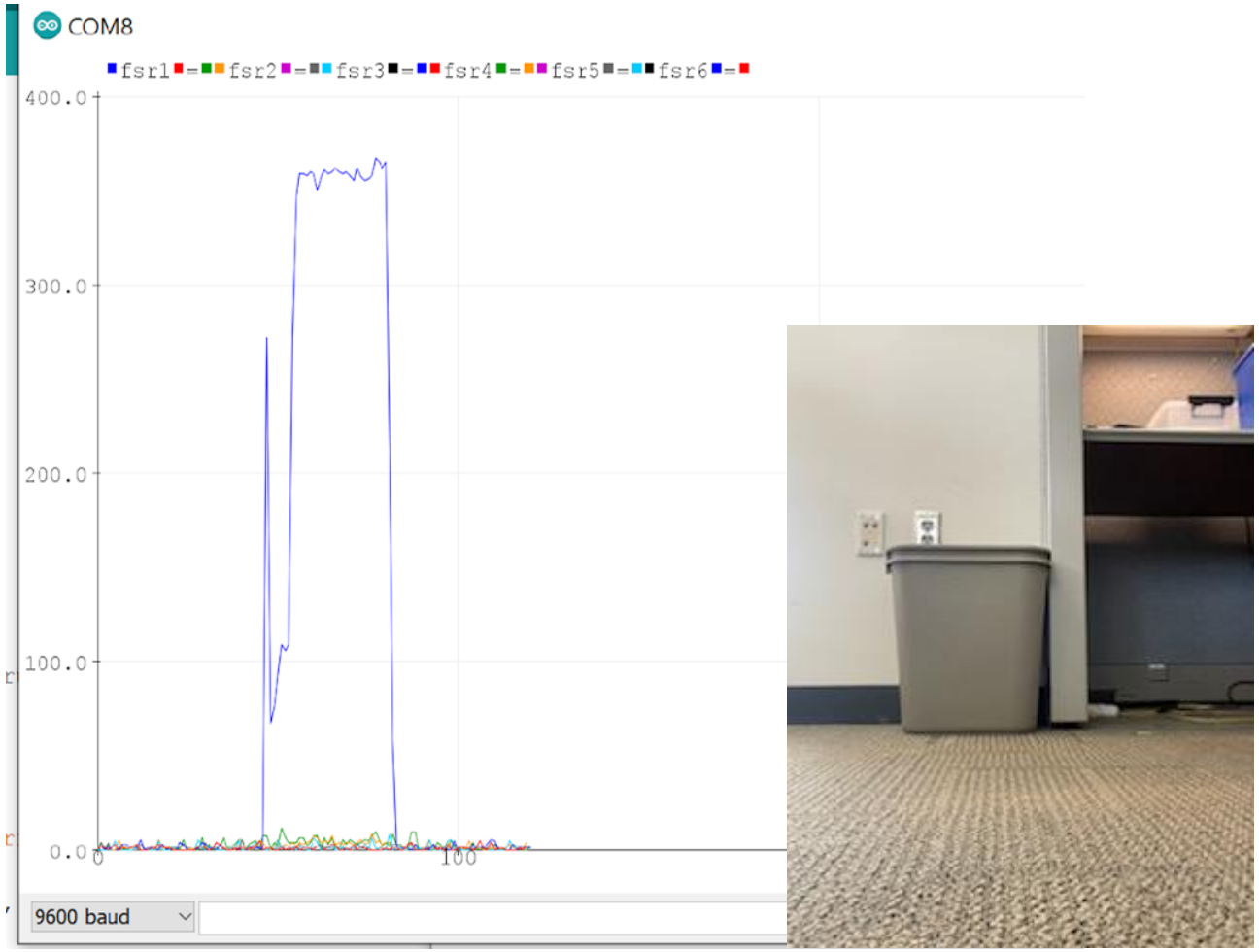
FSR 1
FSR 2
FSR 3
FSR 4
FSR 5
FSR 6


COM8
fsr1 fsr2 fsr3 fsr4 fsr5 fsr6
.6.0
.2.0
8.0
4.0
0.0
0
100
9600 baud



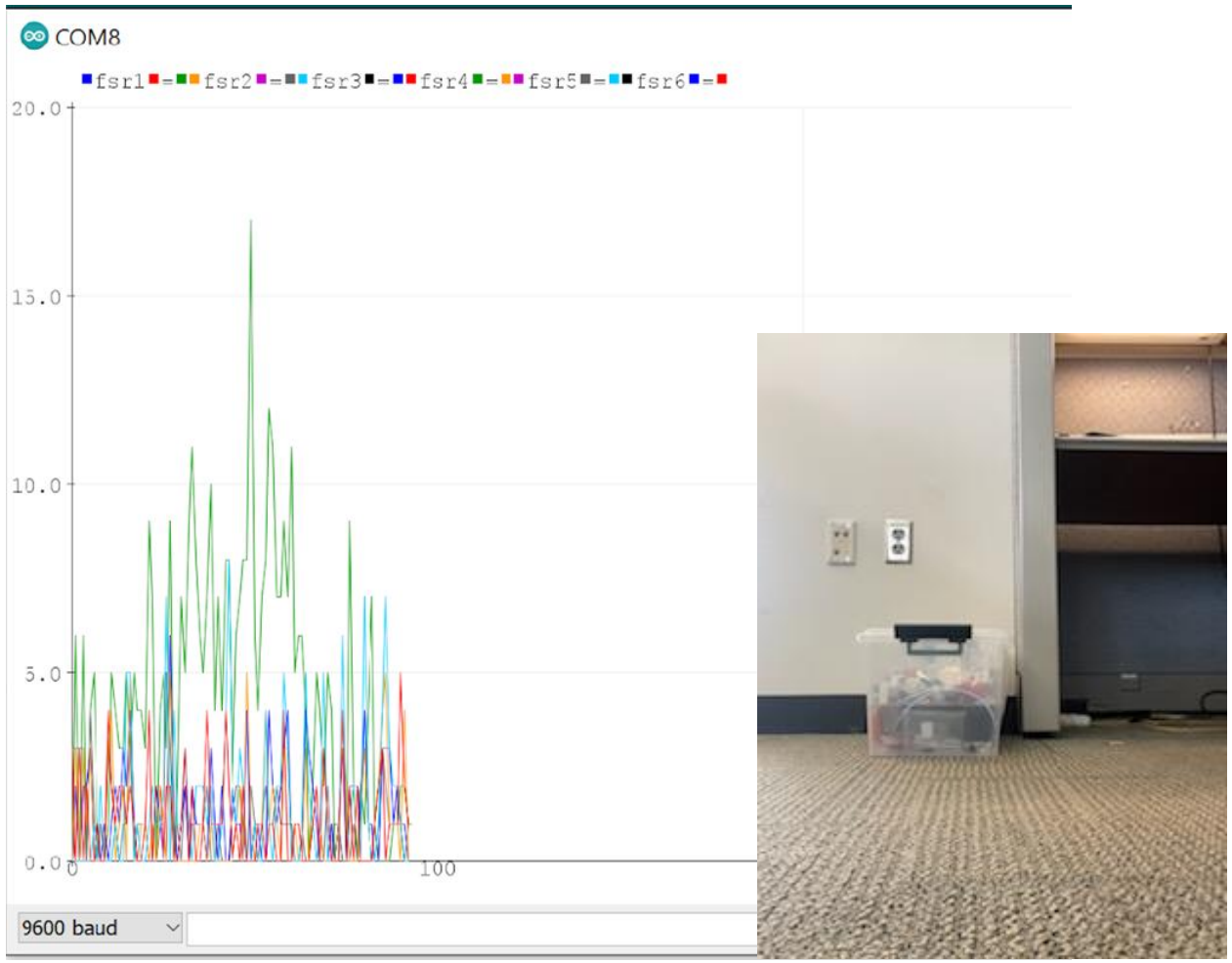
Results

FSR 1
FSR 2
FSR 3
FSR 4
FSR 5
FSR 6


COM8
fsr1 fsr2 fsr3 fsr4 fsr5 fsr6
400.0
300.0
200.0
100.0
0.0
0
100
9600 baud

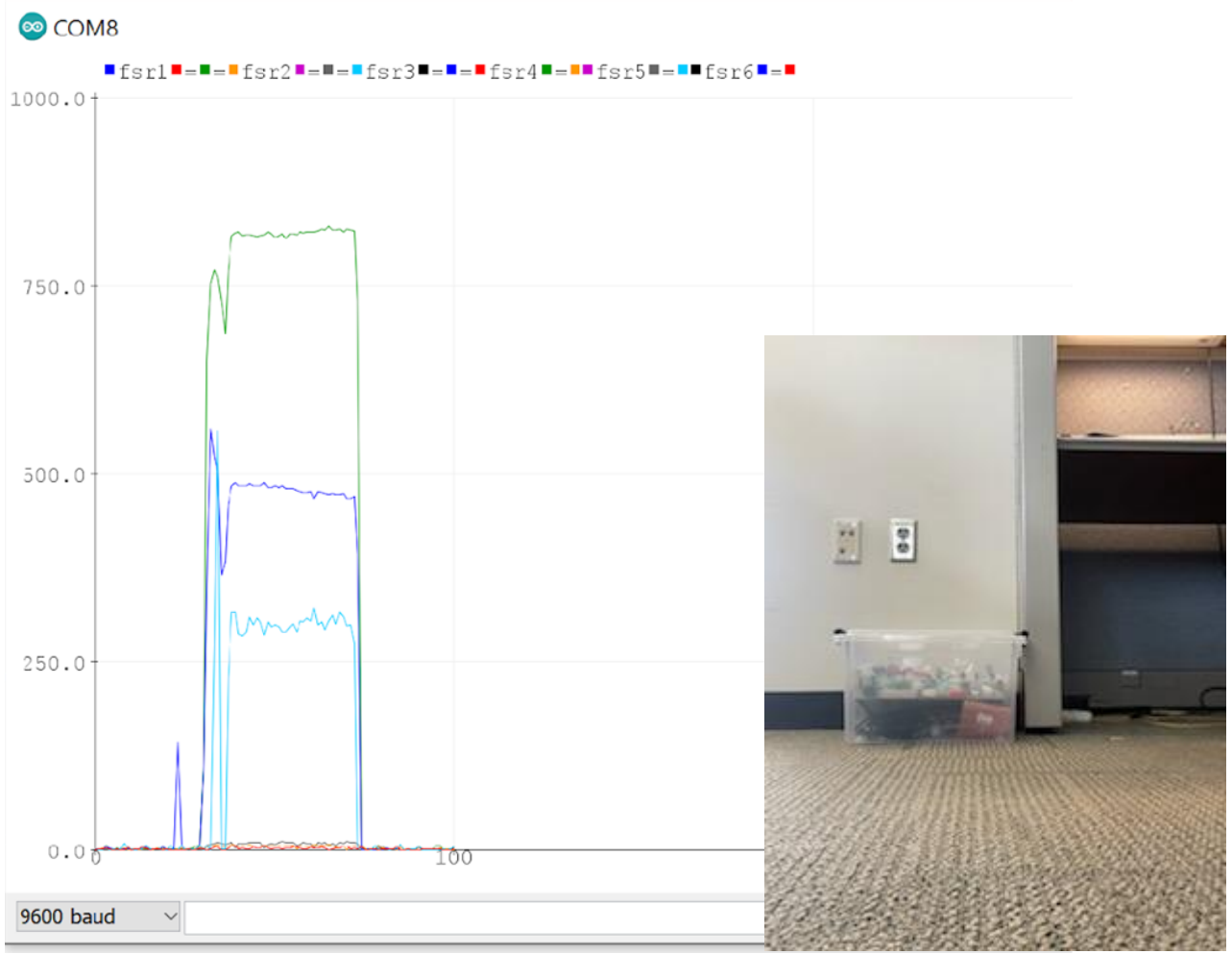
Results

FSR 1
FSR 2
FSR 3
FSR 4
FSR 5
FSR 6


COM8
fsr1 fsr2 fsr3 fsr4 fsr5 fsr6
20.0
15.0
10.0
5.0
0.0
0
100
9600 baud

Results

FSR 1
FSR 2
FSR 3
FSR 4
FSR 5
FSR 6


COM8
fsr1 fsr2 fsr3 fsr4 fsr5 fsr6
1000.0
750.0
500.0
250.0
0.0
0
100
9600 baud

Data Analysis

Small: At rest readings but scaled up

Large: Reading spikes and stays stable

Stiff: Much higher FSR Values

Compliant: Lower FSR Values

Conclusion

Construction of three fingered under actuated grasper

FSRs integrated to silicone grips

Differences between size stiffness recorded

Future Works

- ADDITIONAL AND IMPROVED SENSING
- INCREASED VARIENCE OF STIFFNESS
- FEEDBACK TESTING
- ADDITIONAL SHAPES

Acknowledgements

SURE

Amazon

Intel

Dr. Hammond

ARM Lab

Shane Trent

Thank You!!!